An (Evidence-Based) Agentic Framework for Forecast Error Diagnosis in Global Gradient Boosted Decision Trees

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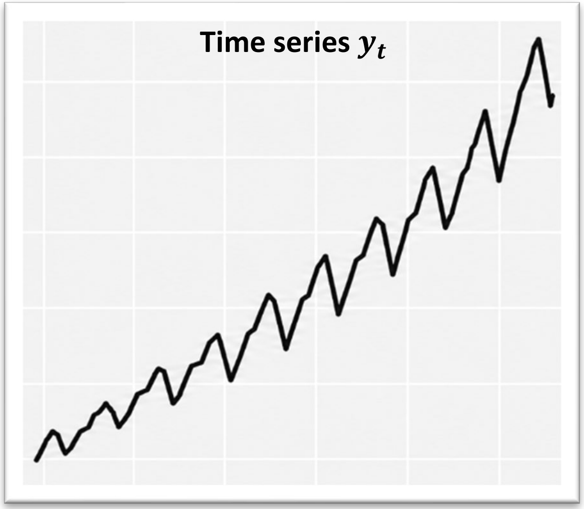
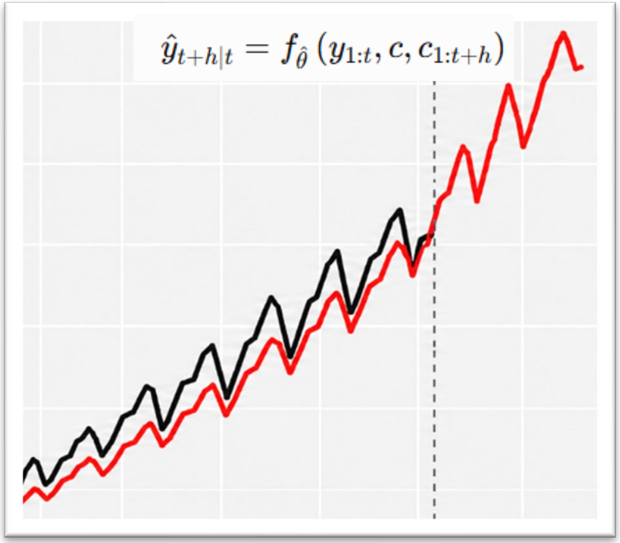
Lets talk about explainability

What exactly are we trying to "explain" here?



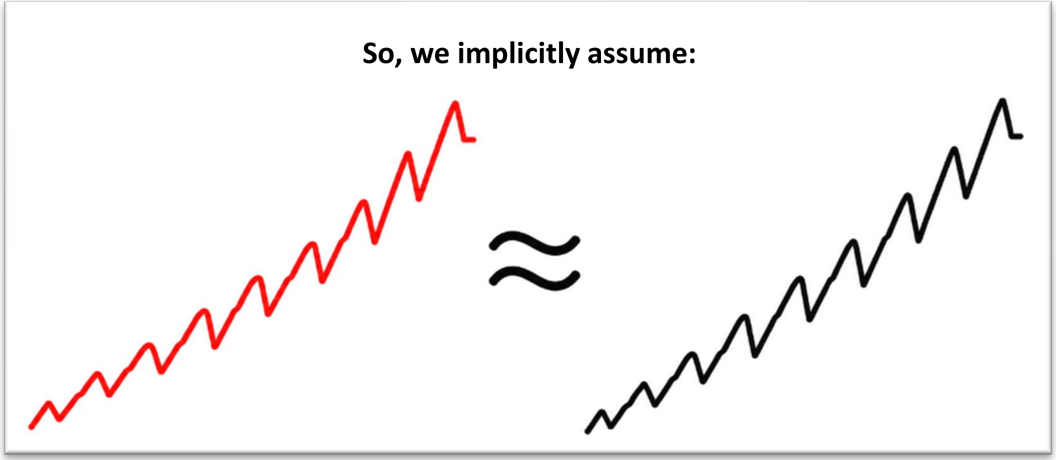
Explain this

Or explain this



(Unless we move from forecast explainability to causal modelling which is a whole other story)

But then we need to make some assumptions.



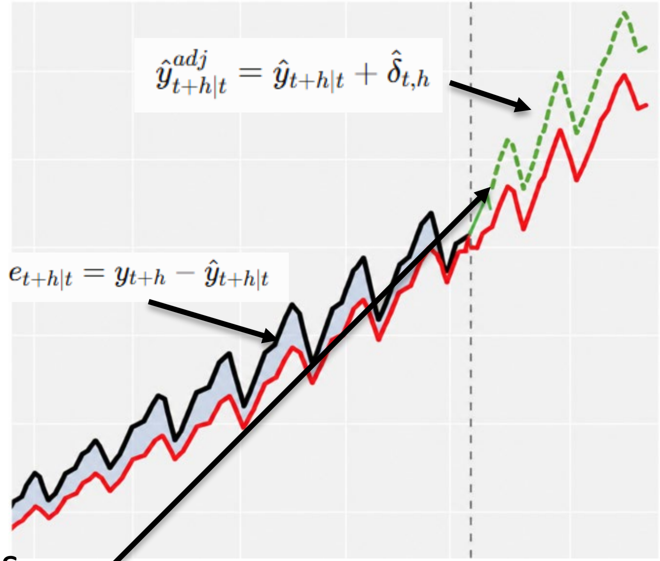
Formally,

- The fitted model has sufficient capacity to capture the predictable structure of the signal.
- The available features are assumed to contain all relevant drivers of variation at the forecast
- The extracted explanation is process-informative, not only model-faithful.
- We trust this explanation blindly as there is no ground truth to score against.

Sagaert, Yves R., and Nikolaos Kourentzes. "Enhancing lasso tactical demand forecasts using semantic information." *International Journal of Production Research* (2025): 1-18.

What if we could reframe the question?

Instead of forecast explanation, we propose error diagnosis
Why the prediction diverged from the realized outcome.



Error diagnosis becomes immediately actionable

- We can retrain if we have data drift
- We can change the model if continuously fails to capture repeated structures (not enough capacity)
- We can include extra features to capture different dynamics
- We can adjust the prediction interval if noise is the main factor.
- Or simply have better feedback for forecast adjustments

Good, now lets about talk Gradient Boosted Decision Trees.

We will focus on Gradient Boosted Decision Trees.

- Empirical evidence that they work (M5, VN, Rohlik sales)
- Battled tests & widely used in production
- We have some evidence on why they work

In our error-diagnosis context:

1. Fast adjustments: Much easier to capture variability within a dataset via feature engineering
2. Partially interpretable (not talking about SHAP)

COMPARATIVE ANALYSIS OF MODERN MACHINE LEARNING MODELS FOR RETAIL SALES FORECASTING

Why do tree-based models still outperform deep learning on tabular data?

When Do Neural Nets Outperform Boosted Trees on Tabular Data?

TABULAR DATA: DEEP LEARNING IS NOT ALL YOU NEED

How Foundational are Foundation Models for Time Series Forecasting?

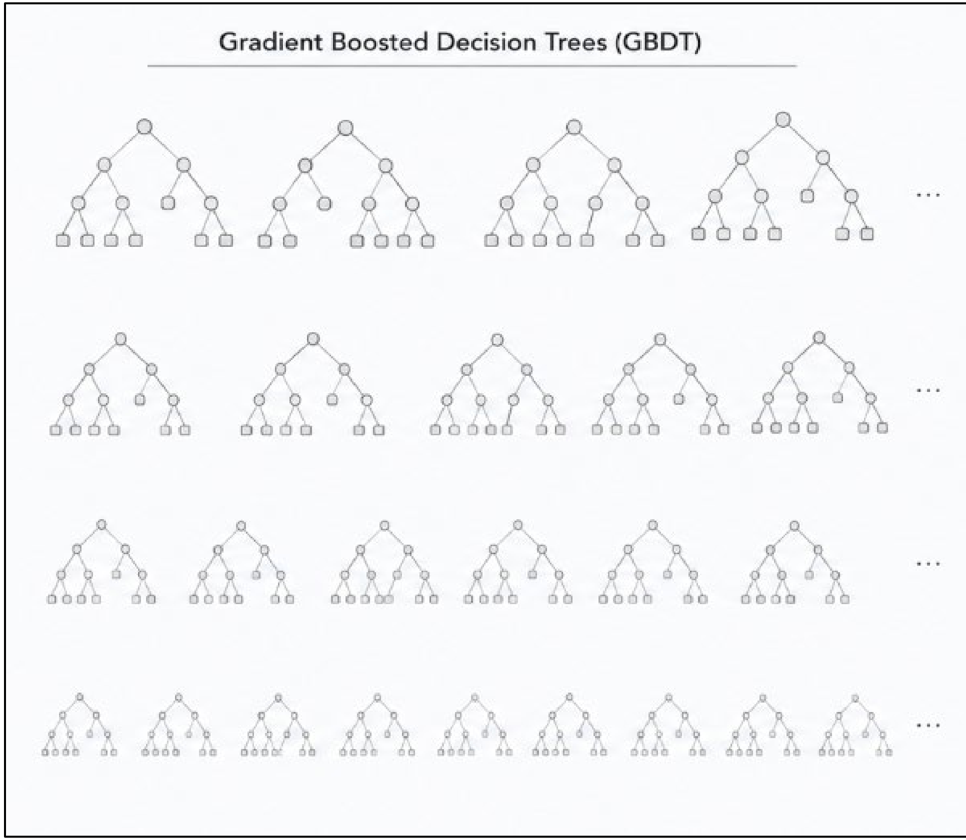
On Identifying Why and When Foundation Models Perform Well on Time-Series Forecasting Using Automated Explanations and Rating

- SHAP values focus on explanations not error diagnosis
- SHAP is model-faithful, not necessarily process-informative or actionable.
- Forecasting features violate independence assumptions.

Theodosiou, F. (2026, January 12). *“In-context learning”, synthetic data and evaluation issues: An (extended) comment on recent advancements in time series foundation models*. Medium.

What partially interpretable means?

A GBDT is an ensemble of n simple estimators

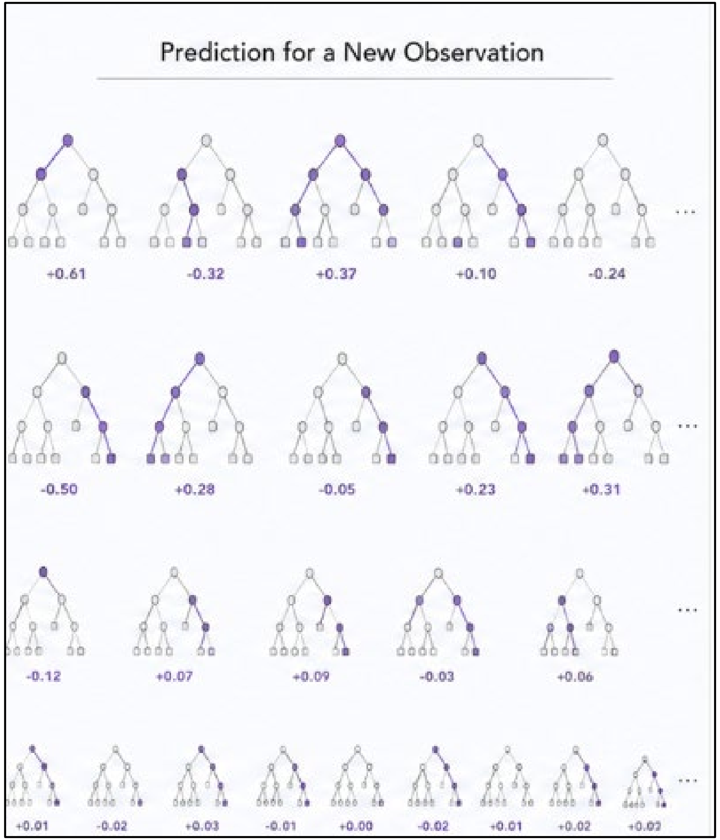


What partially interpretable means?

A GBDT is an ensemble of n simple estimators

- A prediction is the sum of small tree decisions
- Tree decisions threshold are constructed via training.
- For each forecast, we have a deterministic path per tree

  (Feature value -> threshold -> branch -> leaf contribution)



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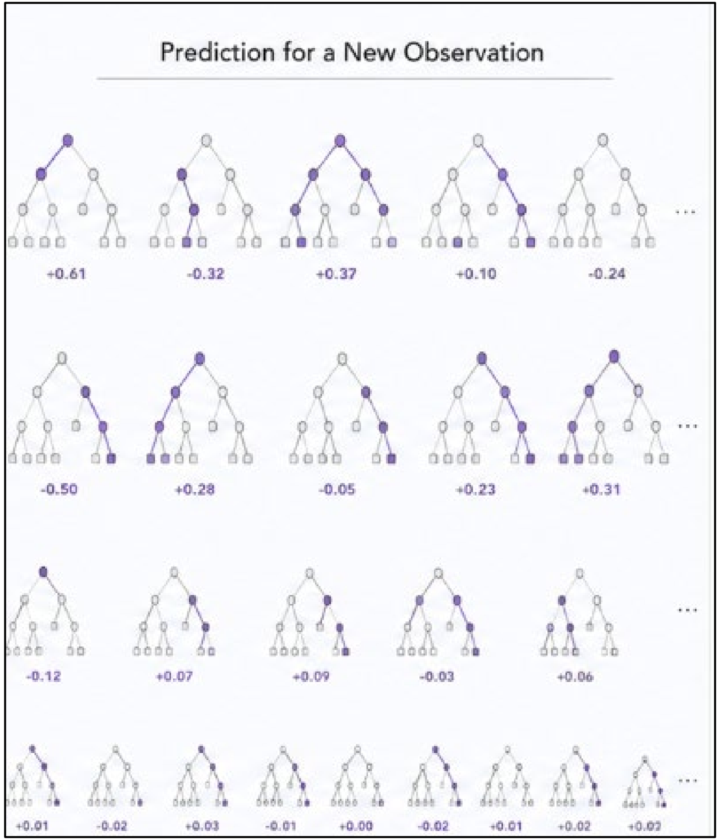
Hypothesis 1:

If learnt tree paths repeatedly appear with similar forecast errors, then these paths might reveal a learnt model-internal mechanism behind the error <u>(model bias)</u>

Hypothesis 2

If a route learnt from the training set fails repeatedly on the validation set, then the model might be applying a pattern that does not generalize <u>(overfitting)</u>



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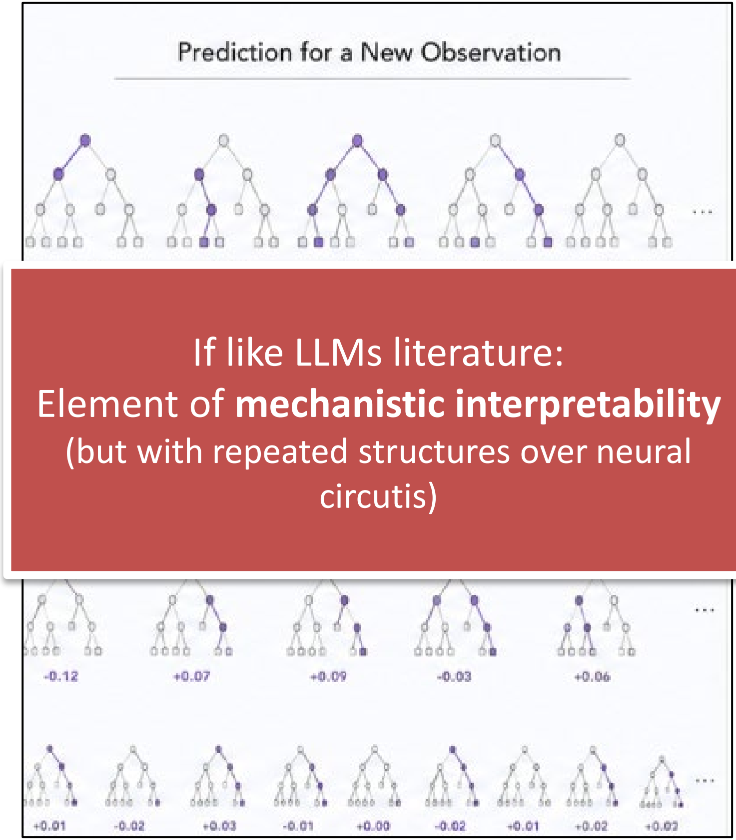


If like LLMs literature:
Element of **mechanistic interpretability**
(but with repeated structures over neural circutis)

Not as straightforward as it sounds. But..

Tree paths are recoverable but not easy to reason over

- A single forecast may activate thousands of splits Relevant evidence can be distributed across trees, horizons, time series
- Repeated structures are not always obvious
- Human inspection become cognitive limiting

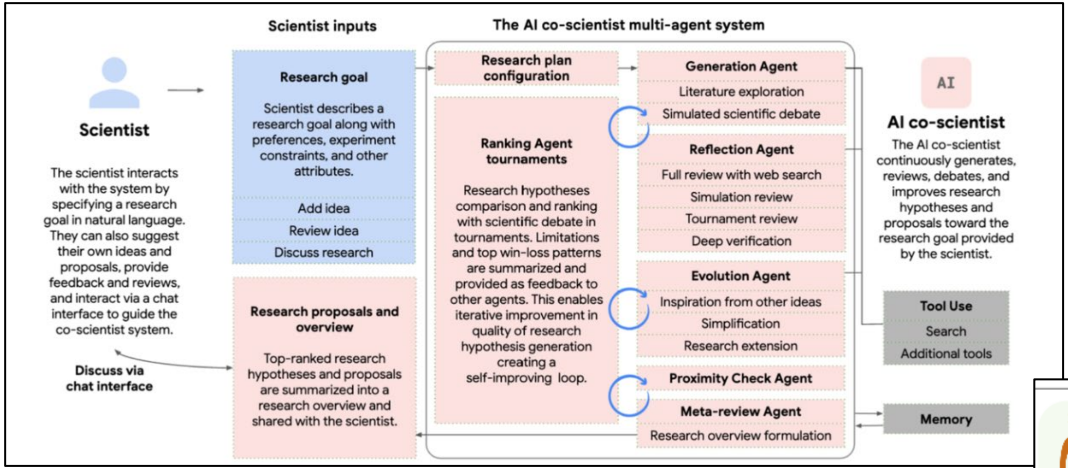
BUT:

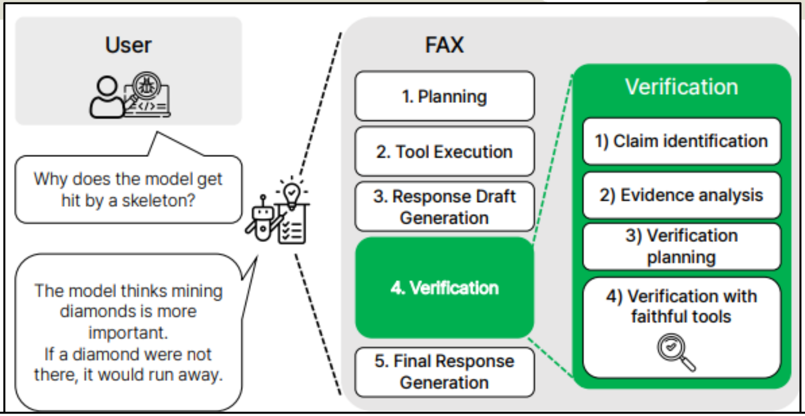
We are in an agentic era.

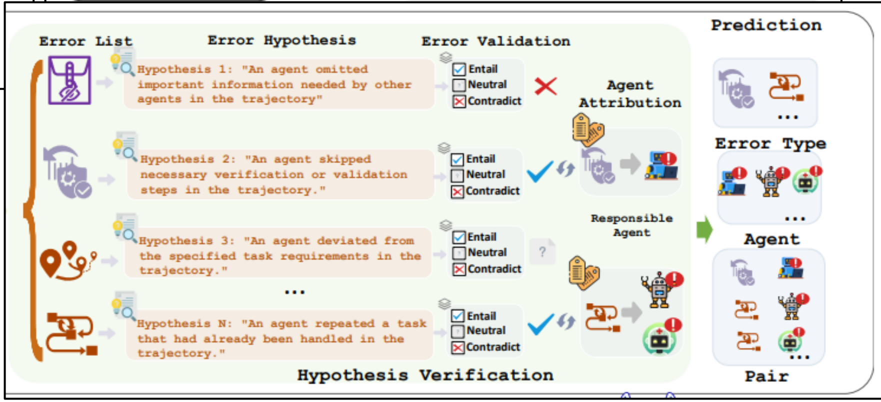
- Long context LLMs are good at searching large evidence sets (eg needle in haystack)
- They can compare many traces and reason over candidate structures
- No cognitive limitations on hypothesis generation
- However, retrieval is not diagnosis.

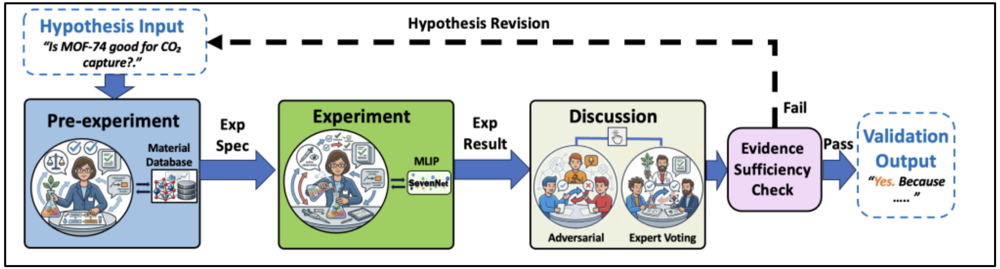
Evidence overload (Forest internals + errors) → Long context search (Retrieve relevant traces – structures) → Hypothesis (Propose candidate error mechanisms) → Verification (Reject/Approve/Refine)

Of course, we did not reinvent the wheel here.









Gottweis, J., Weng, WH., Daryin, A. *et al.* Accelerating scientific discovery with Co-Scientist. *Nature* (2026). https://doi.org/10.1038/s41586-026-10644-y
Kim, Jaechang, et al. "Towards Faithful Agentic XAI: A Verification Method and an Open-World Benchmark for Better Model Faithfulness." *arXiv preprint arXiv:2605.27879* (2026).
Qiao, Hezhe, et al. "VerifyMAS: Hypothesis Verification for Failure Attribution in LLM Multi-Agent Systems." *arXiv preprint arXiv:2605.17467* (2026).
Ahn, Geonhee, et al. "MIND: AI Co-Scientist for Material Research." *arXiv preprint arXiv:2604.13699* (2026).

So, lets put these all together.

Motivation:

- Decision makers care mostly why their model failed and how to adjust
- Explaining a forecast in isolation is not process informative
- Forecasts errors are observable and actionable
- GBDTs give us inspectable internals, LLMs are good at inspecting

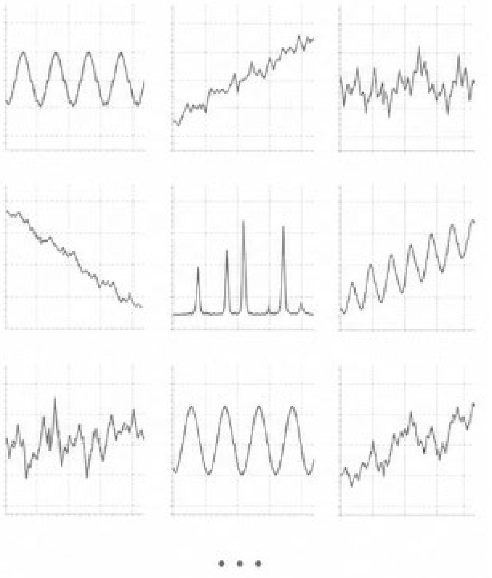
Gap:

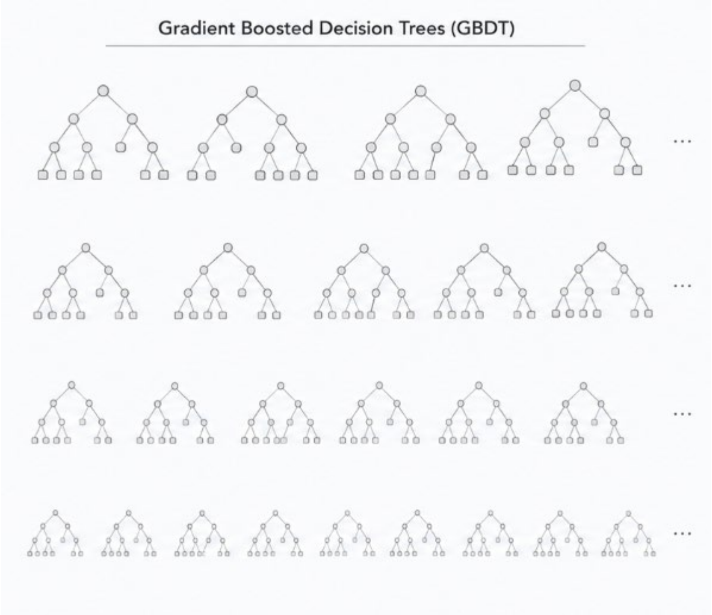
- Most XAI methods are model faithful
- They explain prediction behavior
- They are not measurable.
- GBDTs give us inspectable internals, LLMs are good at inspecting

Our aim:

Build an evidence-grounded agentic framework that uses GBDTs internals to generate, test and validate diagnosis of repeated forecasts errors

How do global GBDTs work?





Pannel of time series

A single unfitted model

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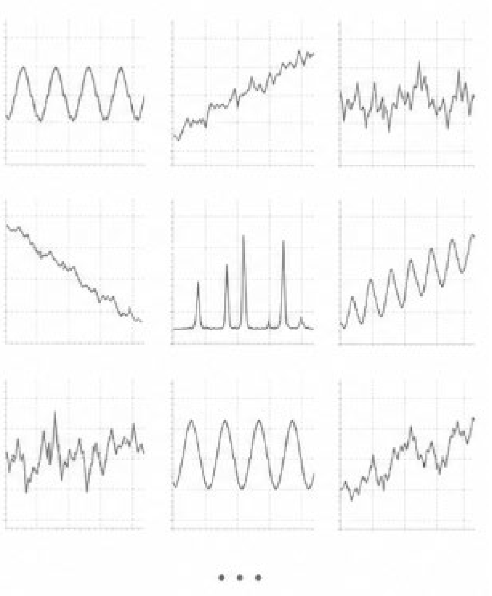

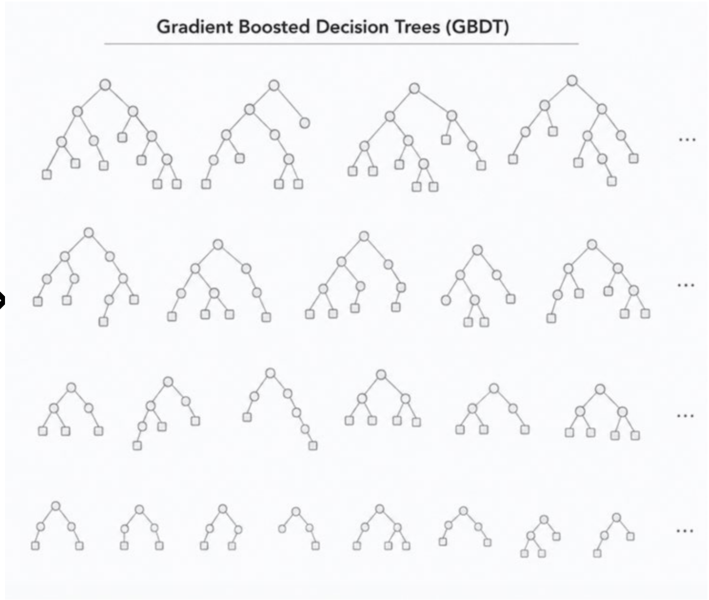

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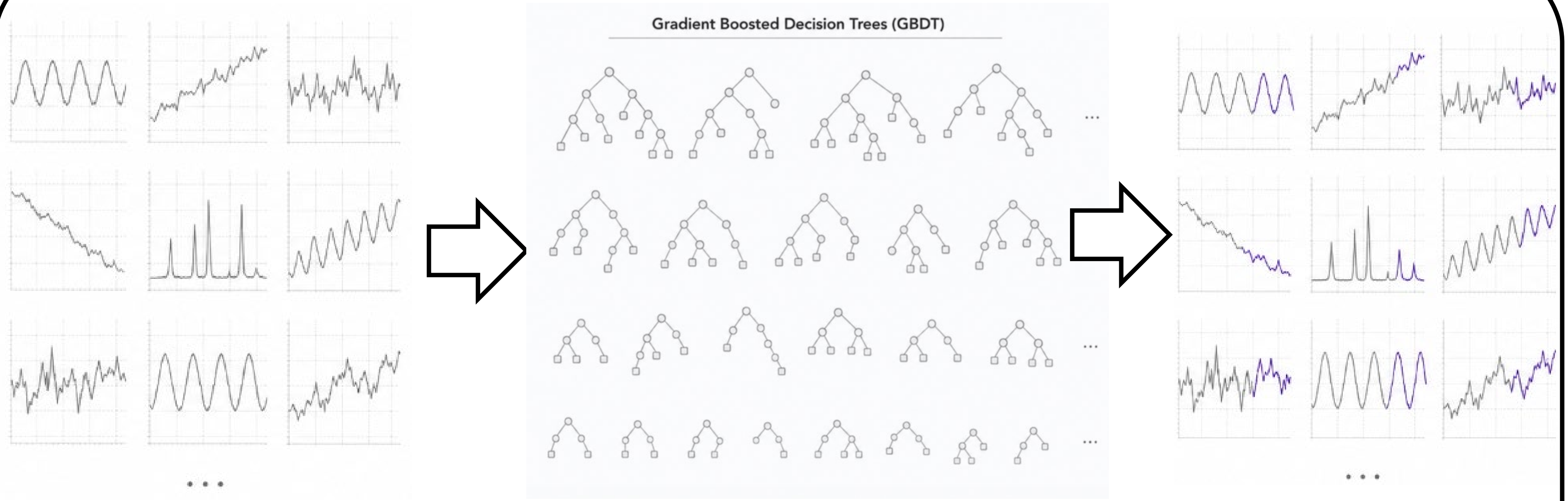


Pannel of time series

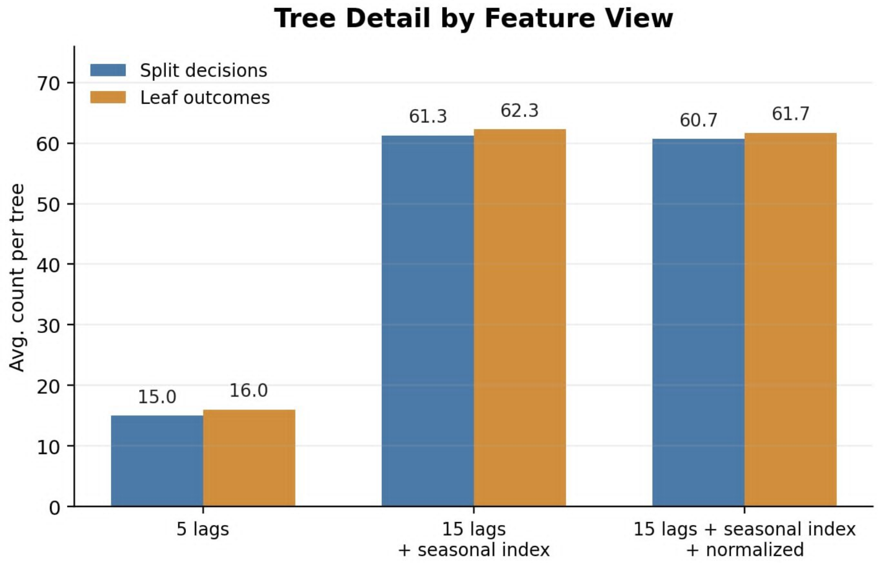
A single fitted model

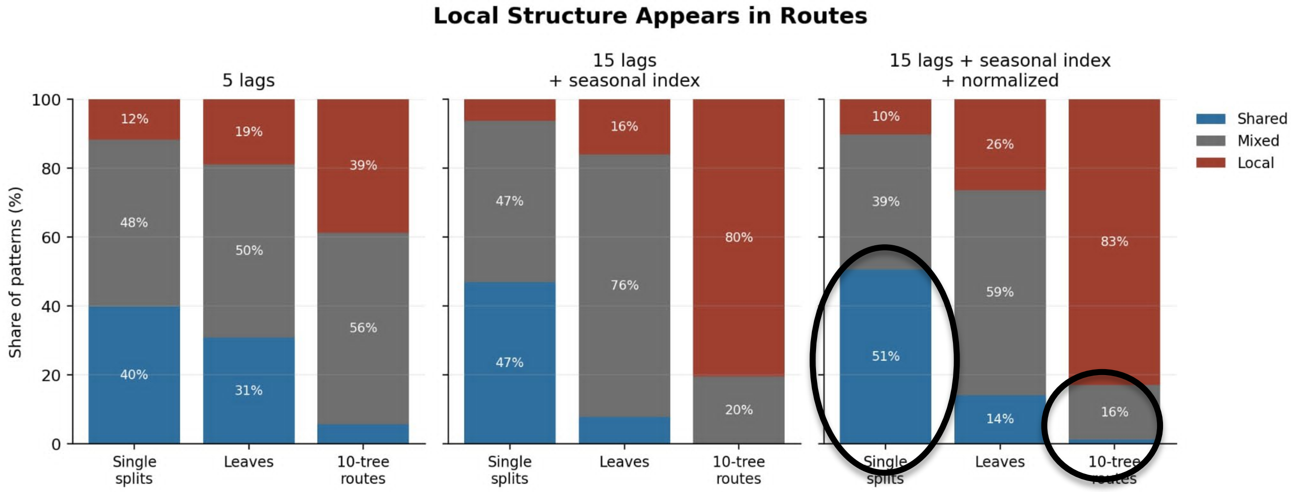
Predictions

Peeking inside the forest:

- Different configurations lead to much different internal structures

- Splits rules are shared, tree routes are local.





Peeking inside the forest:

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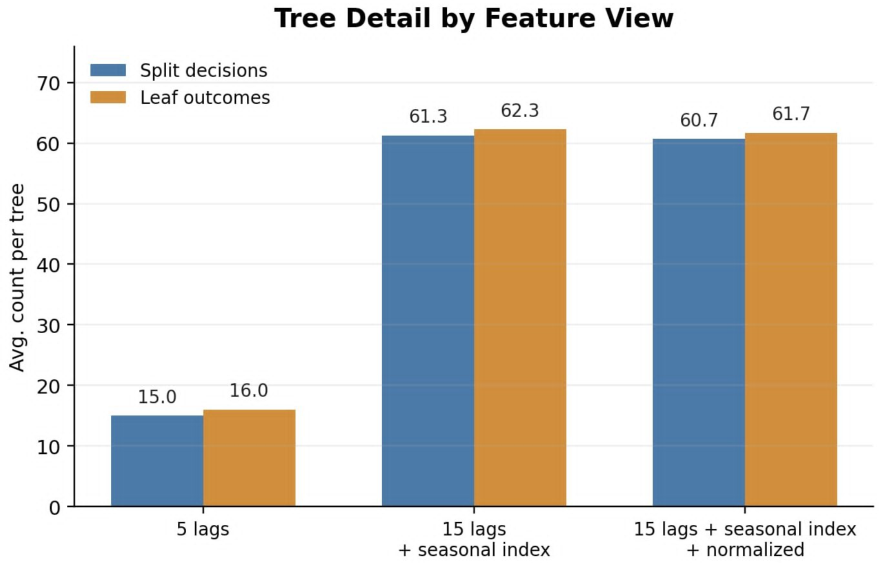
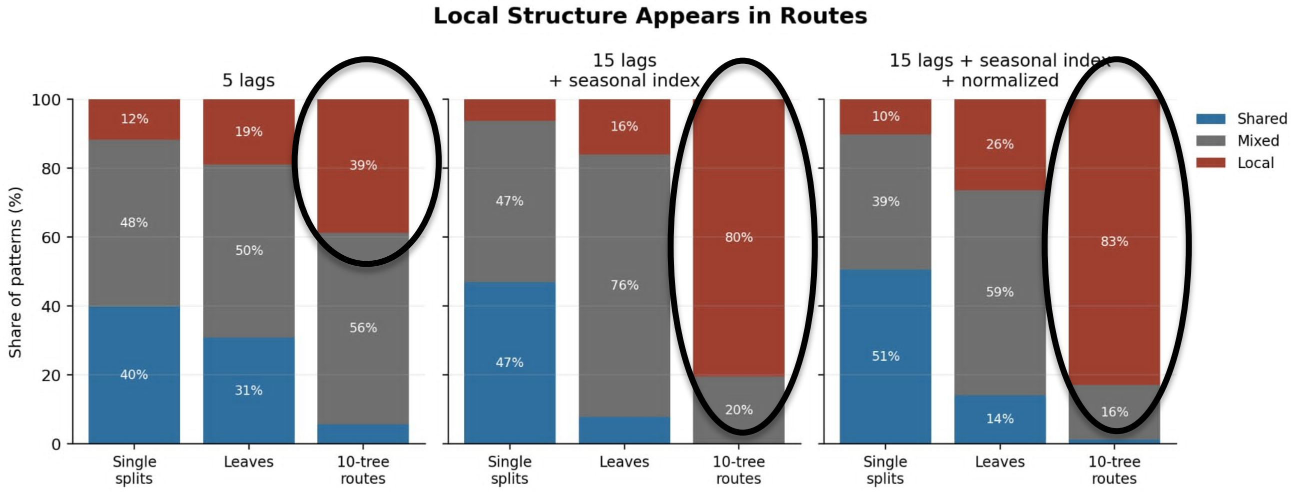
- Simpler models have much more route overlap.

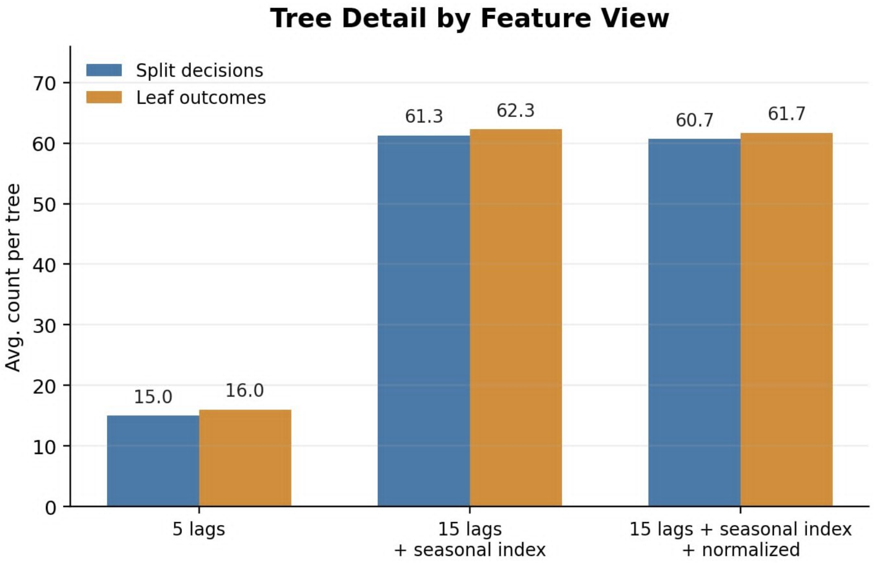


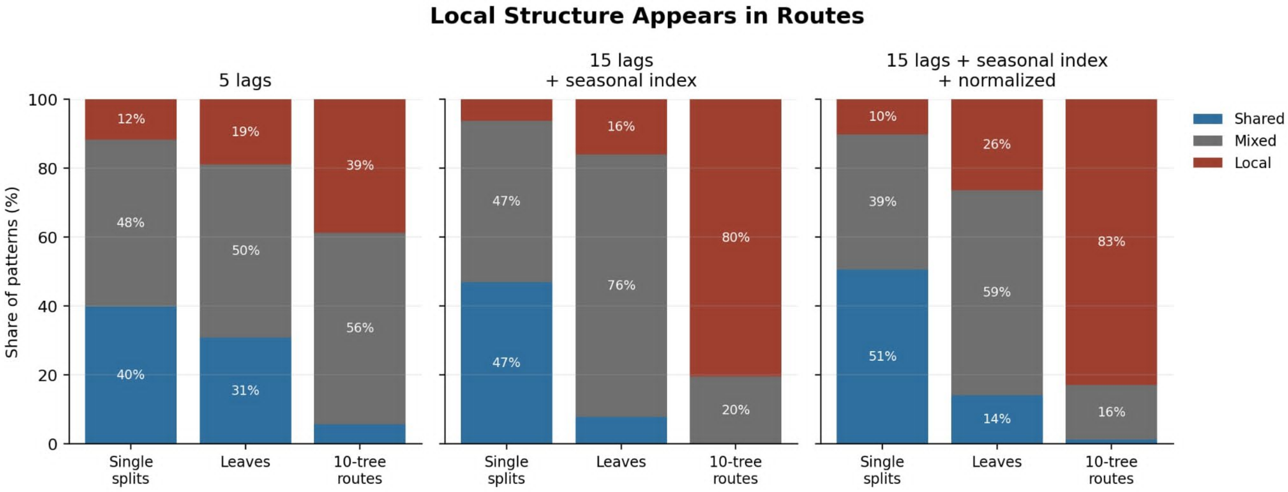


Peeking inside the forest:

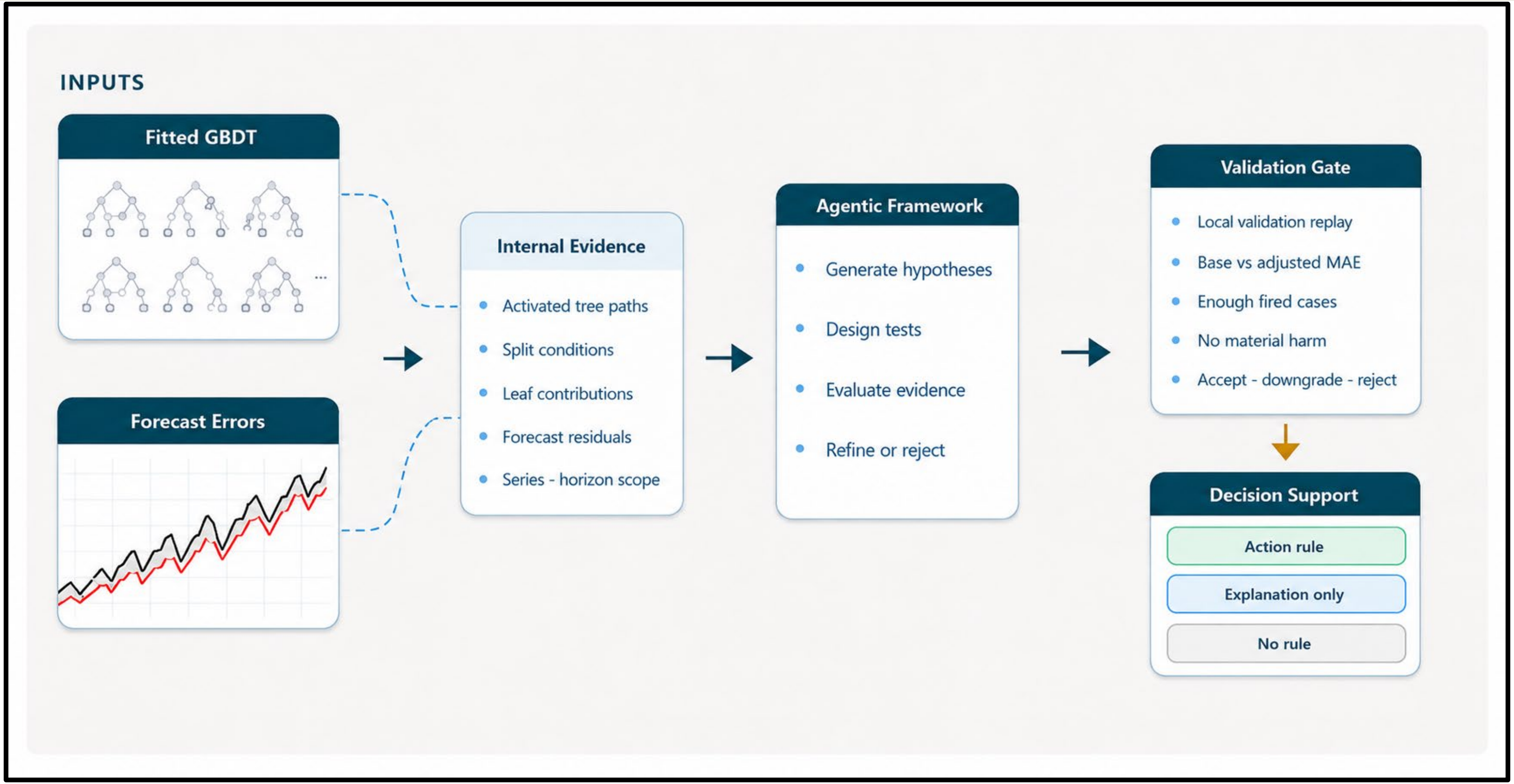
- Different configurations lead to much different internal structures
- Splits rules are shared, tree routes are local.
- Simpler models have much more route overlap.
- No “global” explanations

We need to search how the same splits affect the routes on different time series





Our proposed framework



The agentic framework

Agent 1: LLM selection is very important.

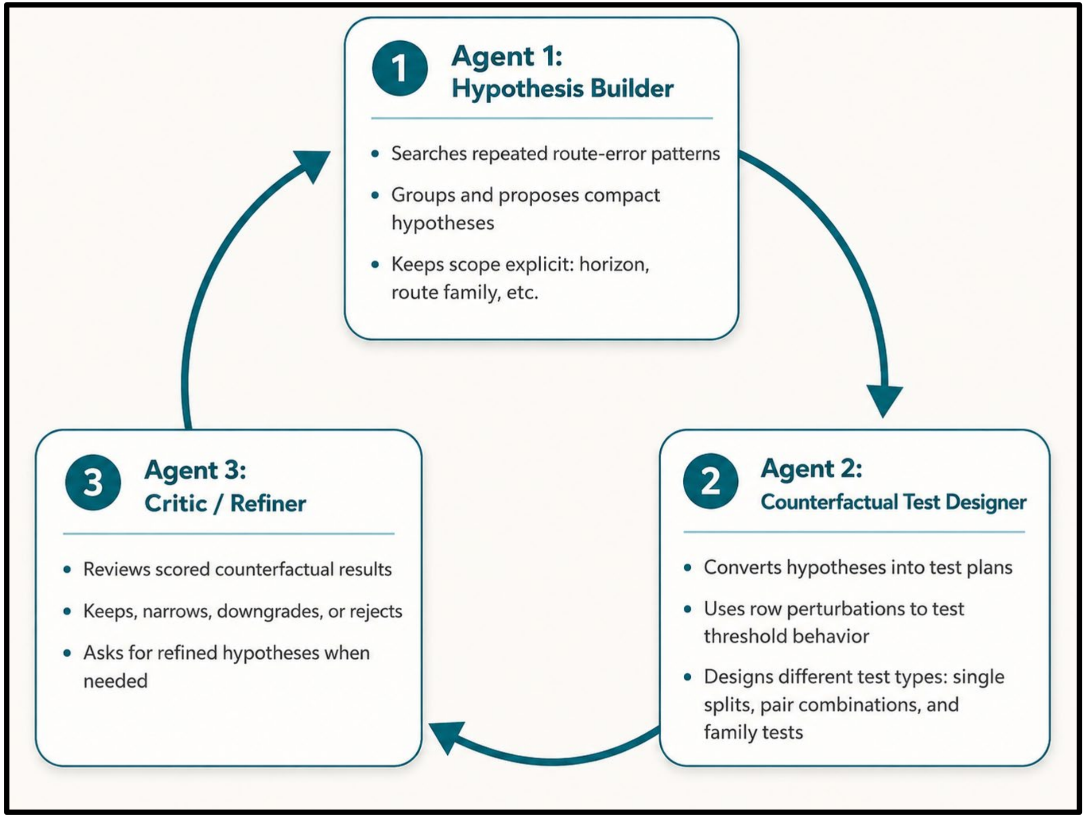
- Reasoning + strict harness are essential
- Hypothesis are constructed hierarchically
- Teacher-student LLM models.

Agent 2: Agent picks the tests.

- Perturbations are changes of the input in controlled ways.
- Usually around the key thresholds.
- Inference with fixed LGBM are cheap

Agent 3: An unbiased judge.

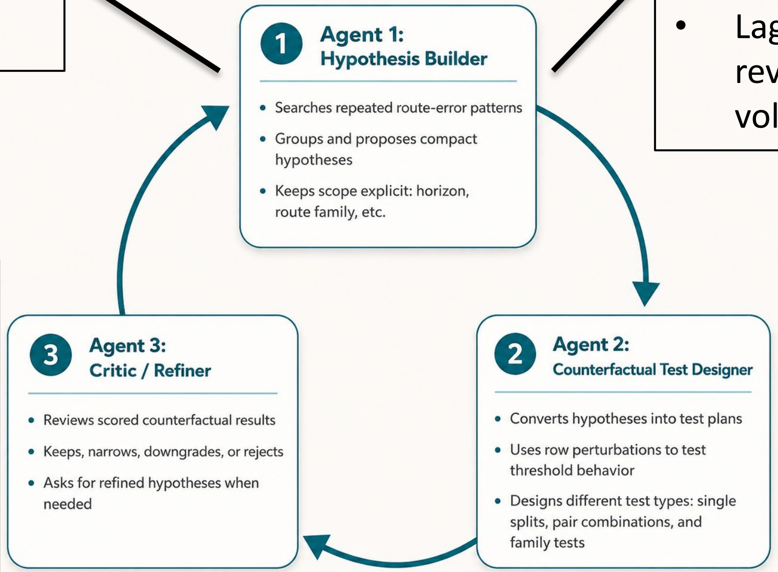
- Agents tend to become biased when evaluating their own feedback
- Picks whether possible adjustments are possible or hypothesis is explanation only



A quick example

Evidence summary:
- Many errors route through low lag5 tree paths.
- Several of these tree also include lag1 and lag2

Hypothesis:
- Lag5 <= 54.5 may identify a general overforecast regime
- Lag5 <= 43 might isolate the route
- Lag5 <= 54.5 AND lag1<= 27.5 might reveal true drops from ordinary low volume periods



Results feedback
- Downgrade lag5<=54.5: Good explanation but too broad
- Keep lag5 <= 43.5: Narrow with life and evidence
- Drop lag1: No clear evidence of added benefit

Counterfactual tests:
- Construct multiple rows with lag5 <=54.5, 52, 50, … 43.5
- Construct rows with adjused lag1 as well (lag5 <= 54.5 and lag1 <=27.5)

A quick experimental design.

Note 1: In contrast with raw explanations, error diagnosis can be evaluated.
Note 2: This is beta version, not yet finalized.

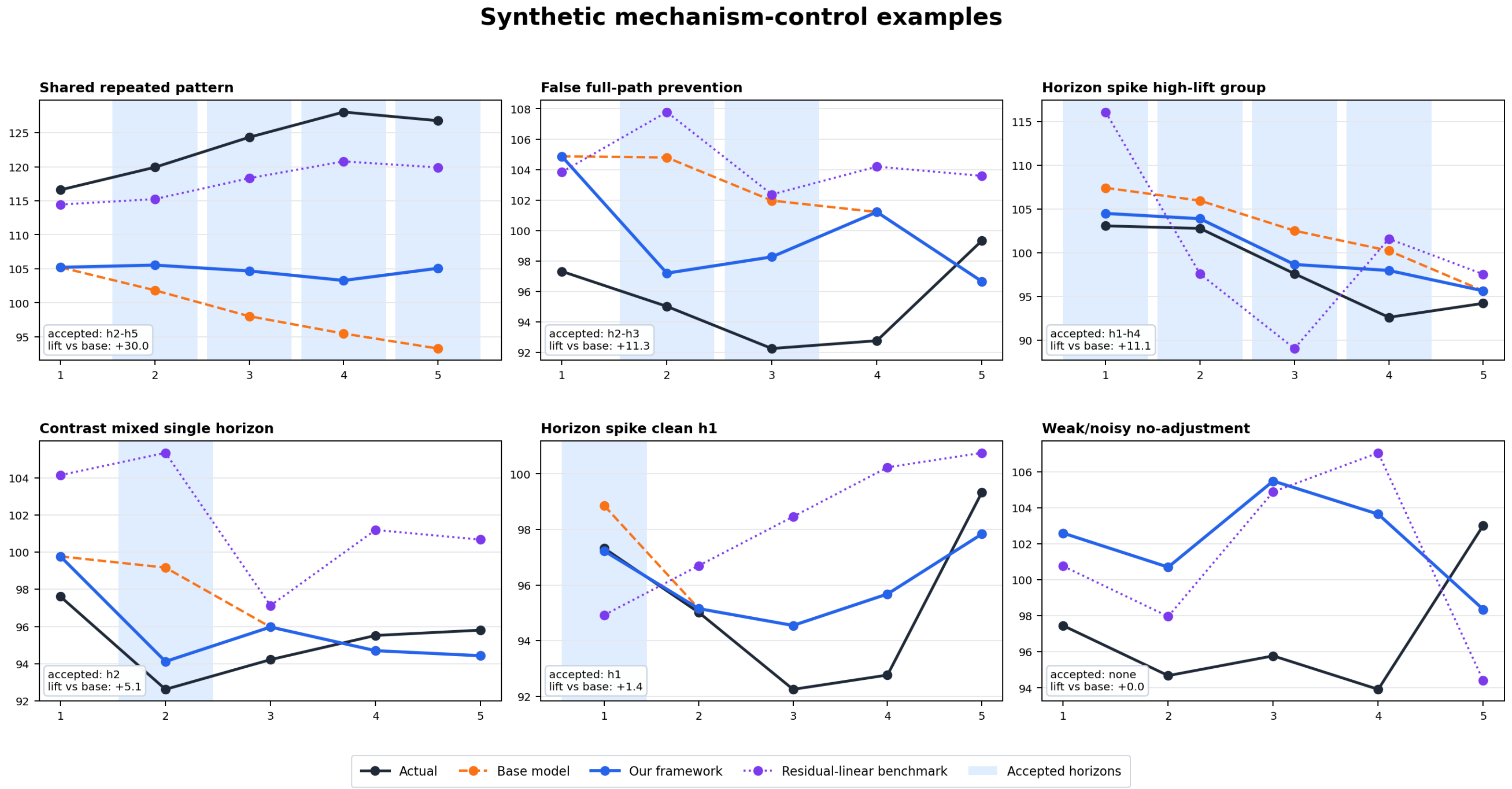
Two types of experiments:

1. Synthetic controlled data -> investigate general behavior
2. Real data from an industry partner:
   - Daily data
   - Ready-to-eat-meal manufacturer
   - 5-day-ahead predictions
   - No covariates/promotions at this point.

Metrics:

- Accuracy lift over base forecast (we use MAE as per our partner preference)
- Adjustment safety: If some triggered adjustments caused performance drop instead of improvement

Synthetic data

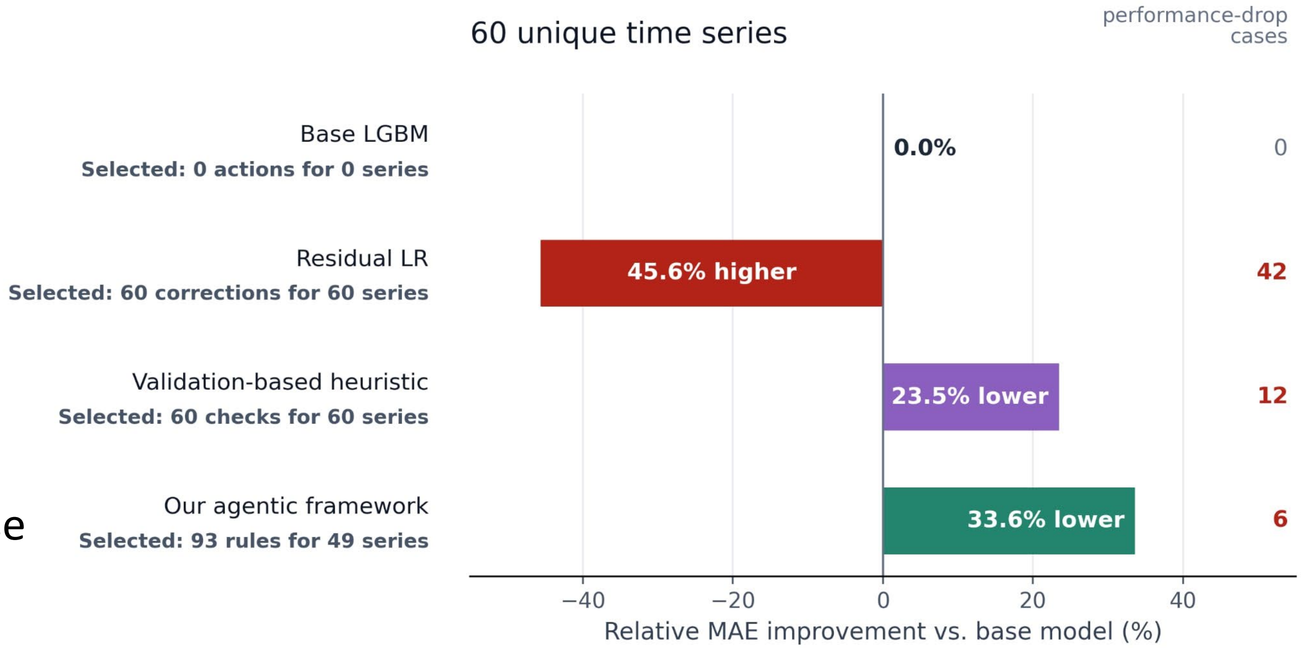


Comparison vs static baselines on real data (OOS)

Note 1: Performance is measured on held-out test sets

Note 2: Adjustments are not proposed for all time series (some do not survive validation)

- The framework proposes adjustments that lead to higher lift over static baselines
- It also proposes less adjustments that lead to performance drop over base

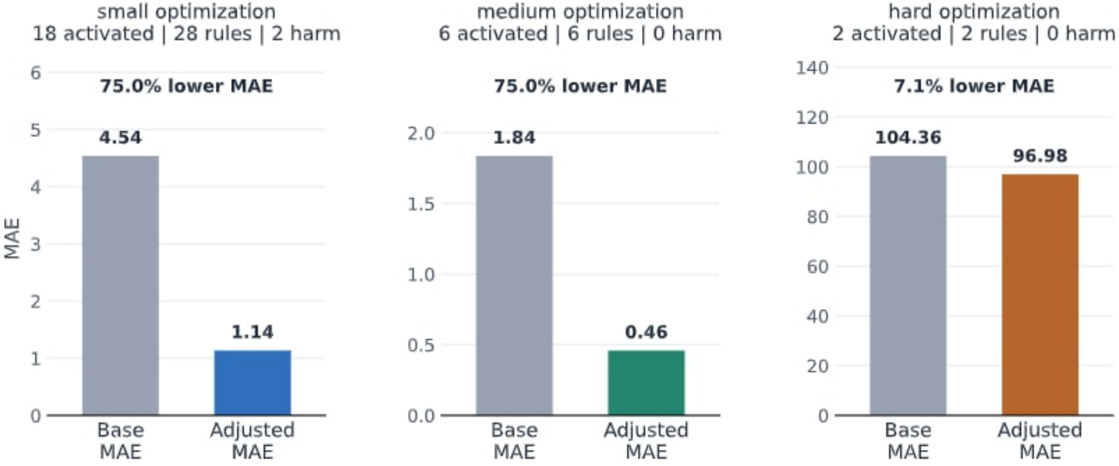


Comparison vs static baselines on real data (OOS)

- Most adjustments are proposed for “simpler” models

- On highly optimized, there are not enough repeated paths to pass validation gate.

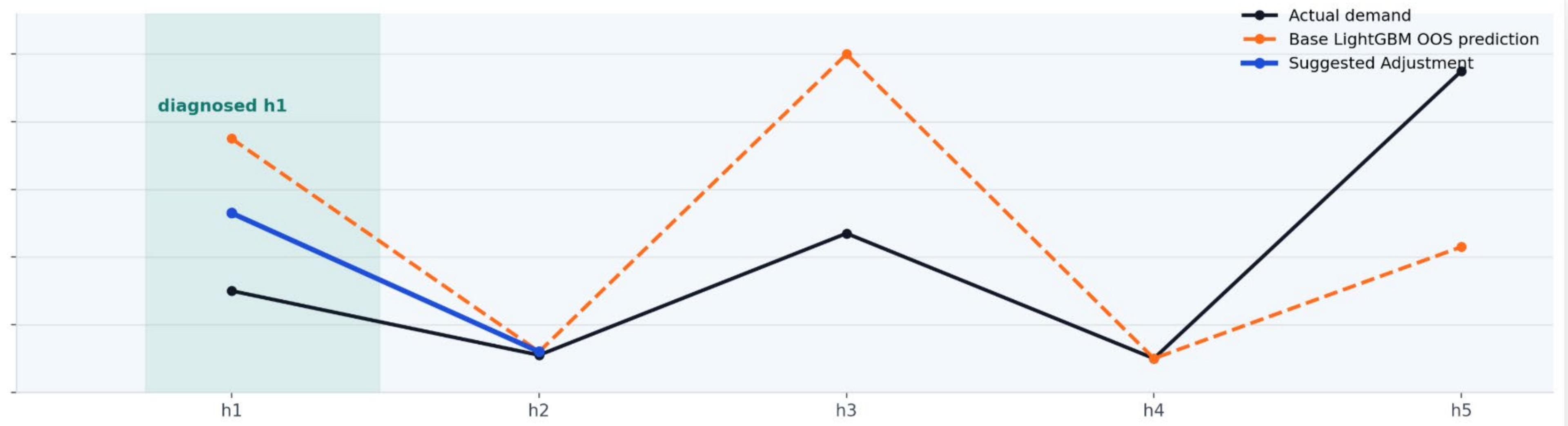
20 unique time series



Example 1: Explanation & Proposed Adjustment

Diagnosis: The base LightGBM made a large first-step overforecast after reading the recent lag pattern as persistent demand. The agents initially saw lag_5 evidence across the whole path, but the counterfactual tests showed that a full-path story was unsafe: it helped some horizons and hurt others. The final diagnosis is horizon-specific: h1 is the forecast step that should be treated as contaminated by the recent-lag route.

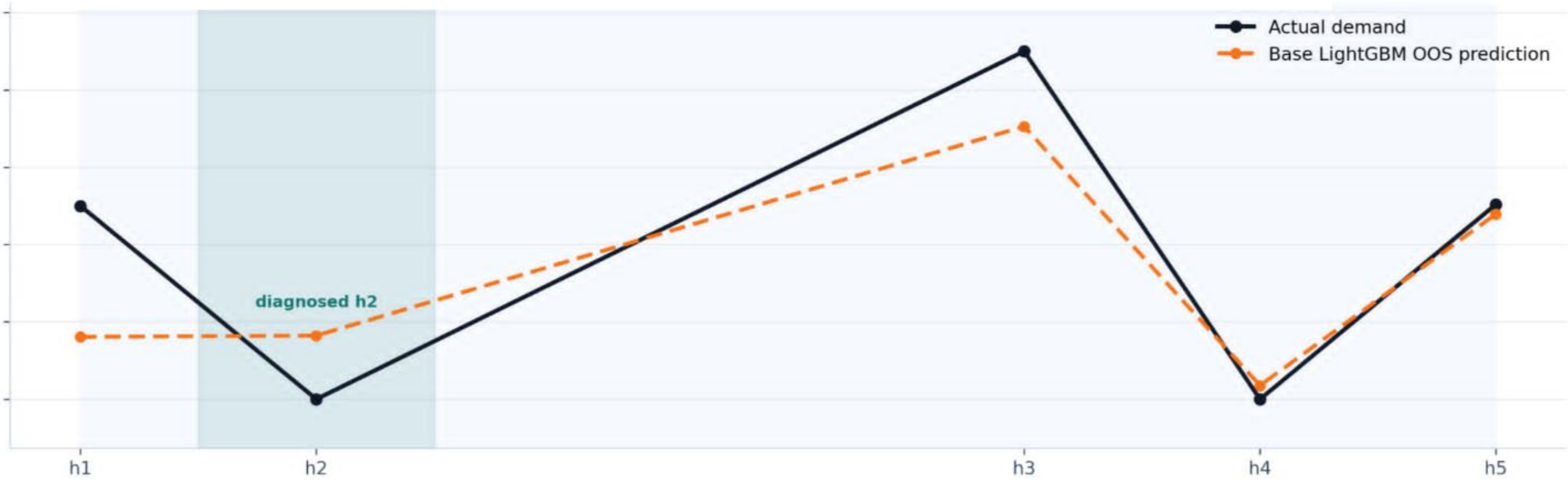
Evidence: lag_5 was repeatedly important, but deterministic sweeps showed the supported correction was narrow. The accepted card kept only h1 instead of promoting a full-path adjustment.



Example 2: Explanation but no adjustment.

Diagnosis: The base path mixes underforecast and overforecast errors, so a single global explanation would be misleading. lag_5 is highly influential across horizons, but its sign alternates. The useful diagnosis is narrower: on h2 the model expects demand where the realized outcome is zero, while other horizons follow a different error pattern.

Evidence: The agent proposed broad lag hypotheses, then deterministic counterfactual tests localized the supported correction to h2. That is the key point: the framework rejects the attractive broad narrative and keeps the evidence-backed horizon.



Conclusions

1. We argue that error diagnosis is more actionable than isolated forecast explanations
2. We propose an agentic framework that analyzes errors for GBDTs and proposes post-forecast adjustments.
3. Early results are promising. Balancing the validation gating is tricky.
4. Potential use cases:
    1. Forecast adjustment proposal (with explanation) for decision makers
    2. Auto-research loop to fit/optimize GBDTs.

Thank you for your attention

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